Embodied Emissions from Manufacture (MTCO2e)	65.37					
Expected Useful Life in Years (Cambridge Systematics)	35	Cambridge Systematics				
Typical Annual Mileage (Cambridge Systematics)	60000	Cambridge Systematics				
Expected Lifetime Mileage	2100000					
Embodied Emissions per Mile Traveled (MTCO2e)	0.0000311307					
DOT-111 Tank Car — Cradle-to-Grave Embodied Emissions						
Materials-Based Quantification						
Charm Industrial — MRV Team | April 2026						
Blue = source-backed | Red = estimate | Purple = GREET2 ref | Black = formula/derived						
1. DOT-111 PHYSICAL SPECIFICATIONS						
Parameter	**Value**	**Unit**	**Source / Reference**	**Classification**		**URL**
Tare weight (light weight)	39,168	kg (86,350 lbs)	GBRX spec sheet (Jan 2021): 29,000-gal DOT 111A100W1, C&I. Quote: "Light weight: 86,350 lbs." Conversion: 86350/2. 20462 = 39168 kg.	SOURCED		https://www.gbrx.com/wp-content/uploads/2021/06/29000-Gallon-DOT-111-Tank-Car.pdf
Configuration	Coiled & insulated (C&I)		GBRX spec sheet. Quote: "coiled and insulated tank car...DOT 111A100W1 specification."	SOURCED		https://www.gbrx.com/wp-content/uploads/2021/06/29000-Gallon-DOT-111-Tank-Car.pdf
Shell material	ASTM A516 Gr 70 carbon steel		49 CFR 179.200-7: permits carbon steel. A516 Gr 70 is industry standard, not regulatory mandate.	PARTIAL		https://www.ecfr.gov/current/title-49/part-179/subpart-D
Tank outer diameter — feet component (GBRX: 9'-10 3/4")	9	feet	GBRX spec: "Tank, outside diameter: 9'-10 3/4"." Feet component = 9.	SOURCED		https://www.gbrx.com/wp-content/uploads/2021/06/29000-Gallon-DOT-111-Tank-Car.pdf
Tank outer diameter — inches component (GBRX: 10 3/4")	10.75	inches	GBRX spec: inches component = 10 3/4 = 10.75 in.	SOURCED		
Tank outer diameter — total inches	118.75	inches	DERIVED. =B11*12 + B12.	DERIVED		
Tank outer diameter — meters	3.0163	m	DERIVED. =B13 x 0.0254 m/in.	DERIVED		
Tank outside length — feet component (GBRX: 53'-4 7/8")	53	feet	GBRX spec: "Tank, outside length: 53'-4 7/8"." Feet component = 53.	SOURCED		https://www.gbrx.com/wp-content/uploads/2021/06/29000-Gallon-DOT-111-Tank-Car.pdf
Tank outside length — inches component (GBRX: 4 7/8")	4.875	inches	GBRX spec: inches component = 4 7/8 = 4.875 in.	SOURCED		
Tank outside length — total inches	640.875	inches	DERIVED. =B15*12 + B16.	DERIVED		
Tank outside length — meters	16.2782	m	DERIVED. =B17 x 0.0254 m/in.	DERIVED		
Min shell thickness — numerator (49 CFR: 7/16 in)	7	(numerator)	49 CFR 179.201-1: "not less than 7/16 inch." Numerator = 7.	SOURCED		https://www.ecfr.gov/current/title-49/part-179/subpart-D
Min shell thickness — denominator	16	(denominator)	49 CFR 179.201-1: denominator = 16.	SOURCED		
Min shell thickness — meters	0.01111	m	DERIVED. =(B19/B20) x 0.0254 m/in.	DERIVED		
Jacket thickness — raw (49 CFR: 11 gauge = 0.1196 in)	0.1196	inches	49 CFR 179.200-4: "not less than 11 gauge (0.1196 inch) nominal." 0.1196 in is the regulatory value.	SOURCED		https://www.govregs.com/regulations/expand/title49_chapterI-i1_part179_subpartD_section179.200-4
Jacket thickness — meters	0.00304	m	DERIVED. =B22 x 0.0254 m/in.	DERIVED		
Steel density (A516 Gr 70)	7,850	kg/m³	AZoM / SSAB: ASTM A516 Gr 70 density 7,850 kg/m³. Standard nominal for carbon steel plate.	SOURCED		https://www.azom.com/article.aspx?ArticleID=4787
Useful service life	50	years	AAR Interchange Rule 88 / OT-24: 50-year condemning limit.	SOURCED		https://public.railinc.com/sites/default/files/documents/OT-24.pdf
2. NON-STEEL MATERIAL INPUTS & GEOMETRY						

Input Parameter	Value	Unit	Source / Reference	Classification		URL
Insulation thickness	0.1016	m (4 in)	PROXY. 49 CFR 179.200-4 sets max thermal conductance of 0.225 Btu/(hr·ft²·°F) but not thickness. Mineral wool k ≈ 0.24 Btu·in/(hr·ft²·°F) (Engineering Toolbox) gives a theoretical minimum of ~1.07 in. UTLX tank car build specifications list 4 inches of fiberglass/mineral wool as industry standard for insulated cars, consistent with FRA-approved thermal protection systems. 4 in = 101.6 mm = 0.1016 m.	PROXY		https://www.utlx.com/build-a-tank-car/4-jacket.html
Insulation density, mineral wool (est.)	113	kg/m³	ESTIMATE. Knauf EPD (DP7-DP8) states rock mineral wool density range 25–200 kg/m³. 113 = midpoint. No railcar-specific density standard identified.	ESTIMATE		https://www.knaufinsulation.com/sites/ki_com/files/uploads/Rock%20Mineral%20Wool%20DP7-DP8.pdf
Insulation overlap/support factor (est.)	1.45		ESTIMATE. No public source. Accounts for batt overlaps, metal support bands, vapor barrier, and installation wastage. 1.45 is an unsourced engineering estimate.	ESTIMATE		
Jacket outer diameter (derived)	3.2195	m	DERIVED. =Tank OD (B14) + 2 x insulation thickness (B29).	DERIVED		
Underframe/truck primer area allowance (est.)	23	m²	ESTIMATE. No public source for tank car underframe surface area. 23 m² is an unsourced engineering estimate based on geometric approximation: 2 longitudinal sills (~18 m x 0.4 m = ~14 m²), transverse members (~4 m²), truck frame surfaces (~4 m²), draft gear area (~1 m²). Immaterial: at 0.5 kg/m² coating rate, this is ~12 kg of coatings, or 0.03% of tare.	ESTIMATE		
Paintable surface area (derived geometry)	204	m²	DERIVED. Jacket exterior cylinder (PI x jacket OD x tank length) + 2 flat-approx heads (2 x PI x (jacket OD/2)²) + underframe allowance (B33).	DERIVED		
Top fittings mass — brass/bronze (est.)	350	kg	ESTIMATE. No public source. Aggregate estimate for manway cover+ring, PRD, safety valve, gauging device, sample/thermo wells, misc nozzle hardware. 350 kg is an unsourced engineering estimate. Immaterial: <1% of tare; even a 50% error changes CTGrave by <0.1 MT.	ESTIMATE		
Coating coverage rate (est.)	0.50	kg/m²	PROXY. Standard industrial coating calculation: DFT x density x 10 = kg/m². For a 2-coat system (primer ~40 µm + topcoat ~50 µm = 90 µm DFT) at epoxy density ~1.2 g/cm³: 90 x 1.2 x 10 = 1.08 kg/m² theoretical. 0.50 kg/m² assumes thinner application typical of rail (less corrosion protection than marine). Conservative: understates coating mass, but coatings are immaterial (<0.3% of tare).	PROXY		
Rubber/elastomers (draft gear cushions, gaskets) (est.)	50	kg	ESTIMATE. No public source. Draft gear elastomer cushion elements + gaskets/seals. 50 kg is an unsourced engineering estimate. Immaterial: 0.1% of tare. Even if doubled, CTGrave impact < 0.01 MT.	ESTIMATE		
Fabrication energy intensity (est.)	2	GJ/t	ESTIMATE. No tank car-specific data. LBNL-62806 REV.2 reports best practice hot rolling at 1.28 GJ/t; UNIDO 2014 benchmarking reports 1.0–1.5 GJ/t for rolling/forming/heat treatment. 2.0 GJ/t adds margin for welding, stress relief, and assembly. Conservative upper bound.	ESTIMATE		https://eta-publications.lbl.gov/sites/default/files/industrial_best_practice_en.pdf

3. MATERIAL MASS BREAKDOWN

Approach: Tare weight is the sourced anchor (GBRX). Non-steel materials are identified and their masses estimated or derived from geometry. Steel mass = tare minus all non-steel. Of the non-steel materials, only thermal insulation is material (~5% of tare). Brass (<1%), coatings (<0.3%), and rubber (0.1%) are marginal.

Material Category	Mass (kg)	% of Tare	Formula / Basis	Classification	EF Applied	
Thermal insulation (mineral wool)	2,654	6.8%	Annular cylinder volume x density x overlap factor. Cylinder only; head insulation (~181 kg) absorbed by steel residual at steel EF — conservative (steel EF > mineral wool EF).	DERIVED from estimates	Mineral wool	
Brass fittings (top fittings assembly)	350	0.9%	From Sec 2. Marginal: <1% of tare. Even a 50% error changes CTGrave by <0.1 MT.	ESTIMATE	Brass (Cu/Zn)	
Coatings + lining (epoxy/polyurethane)	102	0.3%	Surface area (B34) x coverage rate (B36).	DERIVED from estimates	Coatings	
Rubber/elastomers (draft gear, gaskets)	50	0.1%	From Sec 2. Immaterial: 0.1% of tare. Even if doubled, CTGrave impact < 0.01 MT.	ESTIMATE	Rubber	
SUBTOTAL: Non-steel materials	**3,156**	**8.1%**	Sum of insulation + brass + coatings + rubber.			
STEEL (all grades — residual)	**36,012**	**91.9%**	Tare minus non-steel. Covers: shell/heads, jacket, frame, trucks, couplers, draft gear, brakes, heating coils, all structural steel. One blended US steel EF.	DERIVED (residual)	US blended steel	
TOTAL BOM	**39,168**	**100.0%**				
Residual (tare − BOM)	0		Must be 0. Steel absorbs the difference by construction.			

4. EMISSION FACTORS

4A. GREET2 2025 Reference Values (Argonne National Lab)

Material / Parameter	Raw GREET Value	Raw Unit	Converted EF ($kgCO_2e/kg$)	GREET2 Cell Ref	URL
Virgin Steel (BF-BOF)	1408.8	g/lb	3.105869	Mat_Sum, row 65, col B	https://greet.anl.gov/
Recycled Steel (EAF)	292.68	g/lb	0.645248	Mat_Sum, row 65, col C	https://greet.anl.gov/
Copper	1742.3	g/lb	3.841109	Mat_Sum, row 65, col BE	https://greet.anl.gov/
Zinc	1062.57	g/lb	2.342563	Mat_Sum, row 65, col BF	https://greet.anl.gov/
Rubber (synthetic)	1686.16	g/lb	3.717342	Mat_Sum, row 65, col BU	https://greet.anl.gov/
Steel EoL recycling rate	0.96	(fraction)	0.960000	Mat_Sum, cell B19	https://greet.anl.gov/

4B. Derivation Assumptions

Assumption	Value	Unit	Source	URL
US steel: BF-BOF share	0.33	(fraction)	Hasanbeigi & Springer 2019. US 2018: 33% BF-BOF, 67% EAF. Data is 8 years old; EAF share has likely increased, making this conservative.	https://www.bluegreenalliance.org/wp-content/uploads/2021/04/HowCleanisthe U.S.SteelIndustry.pdf
US steel: EAF share	0.67	(fraction)	1 − BF-BOF share (B64).	
Brass: Cu share (C26000)	0.70	(fraction)	UNS C26000 per ASTM B927: Cu 68.5–71.5%, bal Zn. Nominal 70/30.	https://alloys.copper.org/alloy/C26000
Brass: Zn share	0.30	(fraction)	1 − Cu share (B66).	
US grid EF (eGRID 2023)	0.367	$kgCO_2e/kWh$	EPA eGRID 2023 (Jan 2025). US avg: 0.81 lbs CO_2/kWh. 0.81 / 2.20462 = 0.367 kg.	https://www.epa.gov/egrid
US grid EF (per GJ)	0.101945	tCO_2e/GJ	DERIVED. B68 x 277.78 kWh/GJ / 1000.	

4C. Applied Emission Factors

Material Category	EF ($kgCO_2e/kg$)	Formula	Source Classification	Notes
US blended steel (BF-BOF/EAF weighted)	1.457253	BOF_share*BF-BOF_EF + EAF_share*EAF_EF	DERIVED — GREET2 + Hasanbeigi mix	
Mineral wool insulation	1.250000	ICE v4.1: 1.25	DIRECT — ICE v4.1	
Brass/bronze (Cu-Zn, C26000)	3.391546	Cu_share*Cu_EF + Zn_share*Zn_EF	DERIVED — GREET2 Cu+Zn weighted	
Coatings (waterborne paint)	2.152000	ICE v4.1: 2.152	DIRECT — ICE v2.0	
Rubber (synthetic)	3.717342	GREET2 synthetic rubber	DIRECT — GREET2	
Manufacturing energy (per kg of tare)	0.203891	Fab_GJ_per_t x Grid_tCO2e_per_GJ	ESTIMATE — fab energy is unsourced estimate	

5. CRADLE-TO-GATE EMISSIONS

Material Category	Mass (kg)	EF ($kgCO_2e/kg$)	Emissions ($kgCO_2e$)	Emissions ($MT\ CO_2e$)	% of Total
Steel (all grades)	36,012	1.457253	52,478.2	52.478	80.3%
Thermal insulation	2,654	1.250000	3,317.9	3.318	5.1%
Brass fittings	350	3.391546	1,187.0	1.187	1.8%
Coatings + lining	102	2.152000	219.4	0.219	0.3%
Rubber/elastomers	50	3.717342	185.9	0.186	0.3%
Manufacturing / fabrication energy	39,168	0.203891	7,986.0	7.986	12.2%
SUBTOTAL: Cradle-to-Gate			**65,374.4**	**65.374**	

6. END-OF-LIFE METHOD & CRADLE-TO-GRAVE TOTAL

Parameter	Value	Unit	Notes
EoL allocation method	**Cut-off (recycled content)**		No end-of-life recycling credit. Steel LCA typically shows a net-negative EoL due to high recycling rates (~96%), but computing an avoided-burden credit requires data we do not have (net scrap balance, grade-specific virgin/recycled split, scrap market displacement factors). Rather than fabricate those inputs, we conservatively set the EoL credit to zero. This is the cut-off (recycled content) method: the benefit of recycled content is already partially captured in the blended production EF (67% EAF route).
EoL recycling credit	0.000	$MT\ CO_2e$	Zero under cut-off method.

Lifecycle modules C1-C4	Declared, zero impact		Cut-off method: EoL modules declared per Isometric GHG Accounting v1.0 but contribute zero net emissions. Single EoL at 50-year condemning limit (AAR Rule 88).
CRADLE-TO-GRAVE TOTAL	**65.37**	**MT CO_2e**	Equal to CTGate under cut-off method (no EoL credit or debit).
CRADLE-TO-GATE TOTAL	**65.37**	**MT CO_2e**	